

09/04/26

Q1

$$x = \begin{bmatrix} 2 & 4 & 6 \\ 4 & 8 & 12 \\ 6 & 12 & 18 \\ 8 & 16 & 24 \end{bmatrix}$$

Apply PCA & obtain PC equation/s.

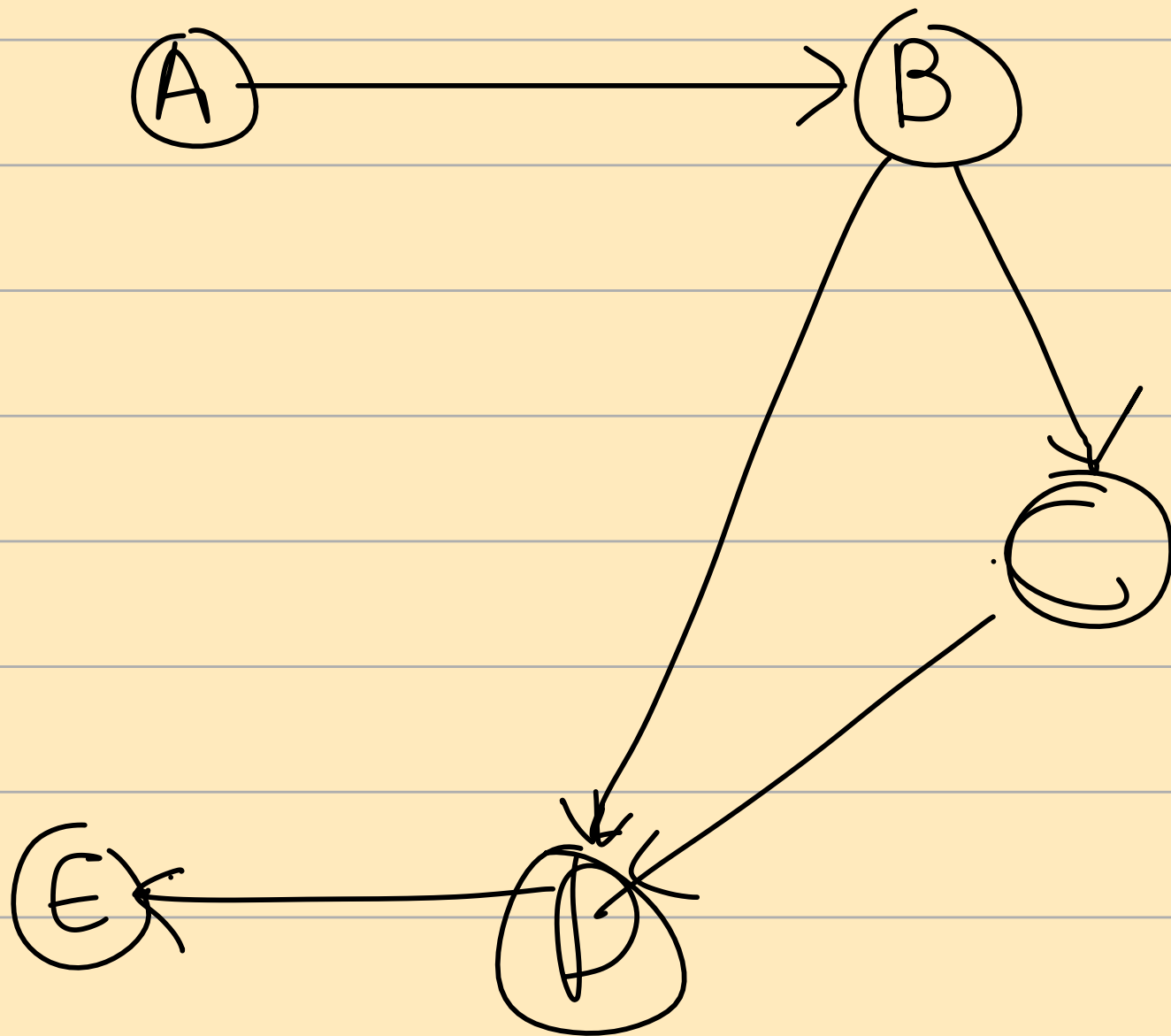
Solⁿ

x_1	x_2	x_3	x'_1	x'_2	x'_3
2	4	6	-3	-6	-9
4	8	12	-1	-2	-3
6	12	18	1	2	3
8	16	24	3	6	9
$\mu_1 = 5$	$\mu_2 = 10$	$\mu_3 = 15$			

$$\text{So, } x_c = \begin{bmatrix} -3 & -6 & -9 \\ -1 & -2 & -3 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{bmatrix}$$

$PC_1 : 0.26x_1 + 0.53x_2 + 0.80x_3$

Graph Theory :



<i><ii> Centrality / Degree

A	:	2
B	:	3
C	:	3
D	:	3
E	:	1

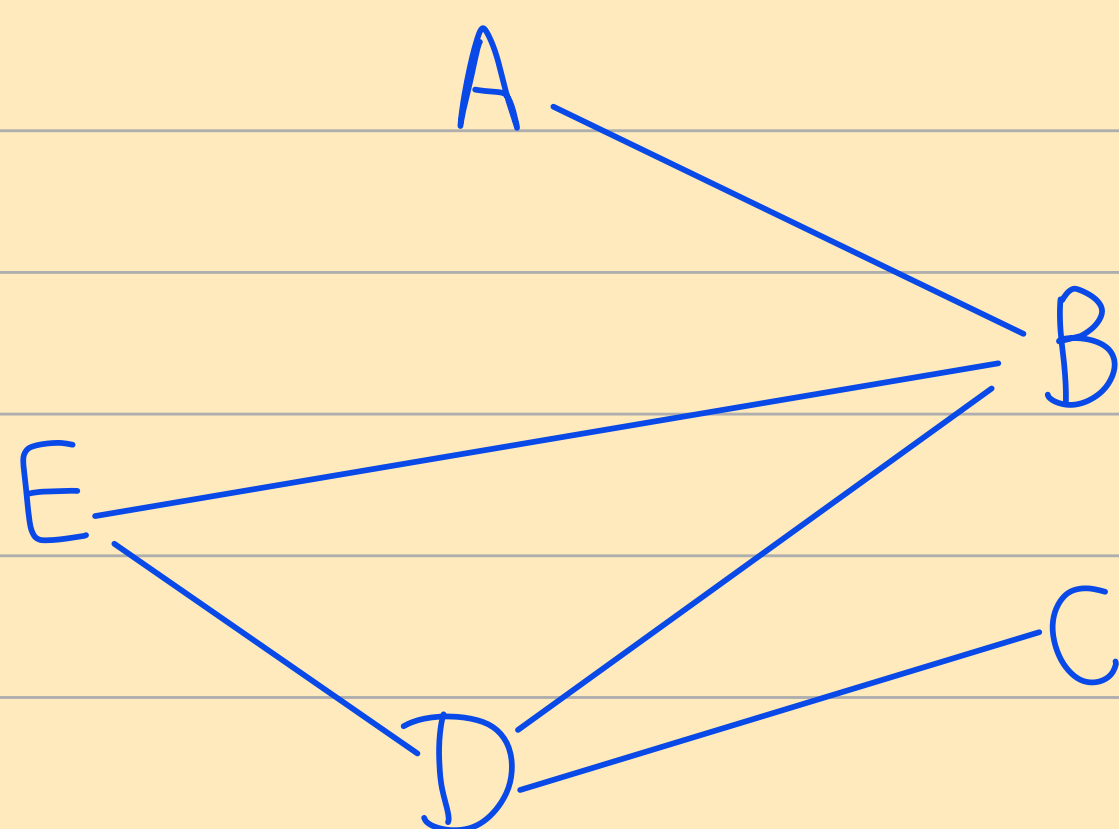
<ii> Betweenness :

• A → E : D

<iii> Closeness :

A

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(i) Degree / Centrality

$$A = 1$$

$$B = 4$$

$$C = 2$$

$$D = 3$$

$$E = 2$$

(ii) Closeness : Closeness of B

$$B \rightarrow A : 1$$

$$B \rightarrow C : 1$$

$$B \rightarrow D : 1$$

$$B \rightarrow E : 1$$

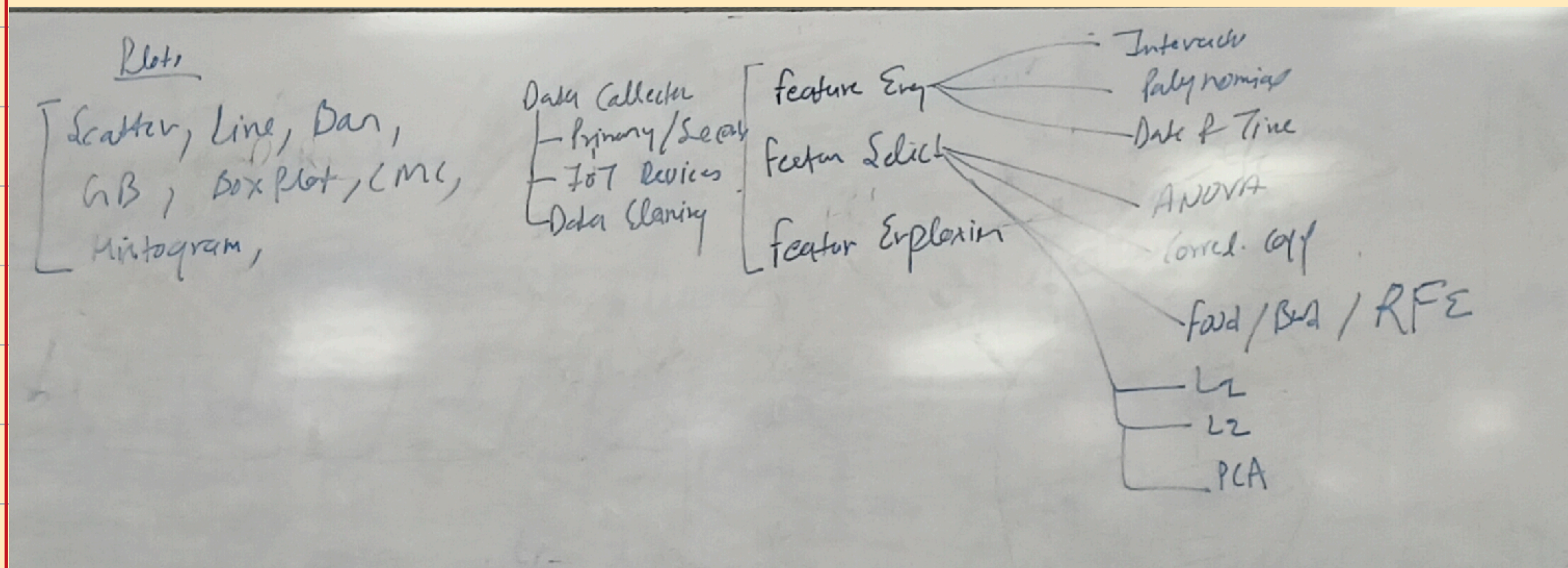
$$\begin{aligned} \text{Closeness} &= \frac{1}{1+1+1+1} \\ &= \frac{1}{4} = 0.25 \end{aligned}$$

(iii) Between / Bridge

A to E

min path : A - B - E

⇒ Syllabus :



16/04/26

Class	Score
A	10, 12
B	20, 22
	$\mu = 16$

Apply ANOVA & find F.Score

$$\mu_A = 11$$

$$\mu_B = 21$$

$$\text{Total mean} = \frac{64}{4} = 16$$

$$F\text{-score} = \frac{\text{Variance b/w groups}}{\text{Variance within groups}}$$

$$\begin{aligned} \text{Variance b/w groups} &= \left[(11-16)^2 + (21-16)^2 \right] \\ &= 50 \end{aligned}$$

$$\begin{aligned} \text{Variance within groups} &= \left[(10-11)^2 + (12-11)^2 \right] + \\ &\quad \left[(20-21)^2 + (22-21)^2 \right] \\ &= 1 + 1 + 1 + 1 \\ &= 4 \end{aligned}$$

$$F\text{-score} = \frac{50}{4} = 12.5$$

Q2

Study hours

Marks

1	2	→ 1.5
2	4	→ 3
3	6	→ 4.5

$$12 + 6 = 18 \quad \frac{18}{6} = 3$$

$$\begin{aligned}\text{Variance b/w} &= (1.5 - 3)^2 + (3 - 3)^2 + (4.5 - 3)^2 \\ &= (1.5)^2 + (1.5)^2 \\ &= 4.5\end{aligned}$$

$$\begin{aligned}\text{Variance within} &= (1 - 1.5)^2 + (2 - 1.5)^2 \\ &\quad + (2 - 3)^2 + (4 - 3)^2 + \\ &\quad (3 - 4.5)^2 + (6 - 4.5)^2 \\ &= (0.5)^2 * 2 + (1) + (1) + \\ &\quad (-1.5)^2 * 2 \\ &= 0.5 + 2 + 4.5 \\ &= 7\end{aligned}$$

$$\frac{4.5}{7} = 0.64$$